HW 8 - Lagrange Basis Function Derivatives

Jordan Sheppard

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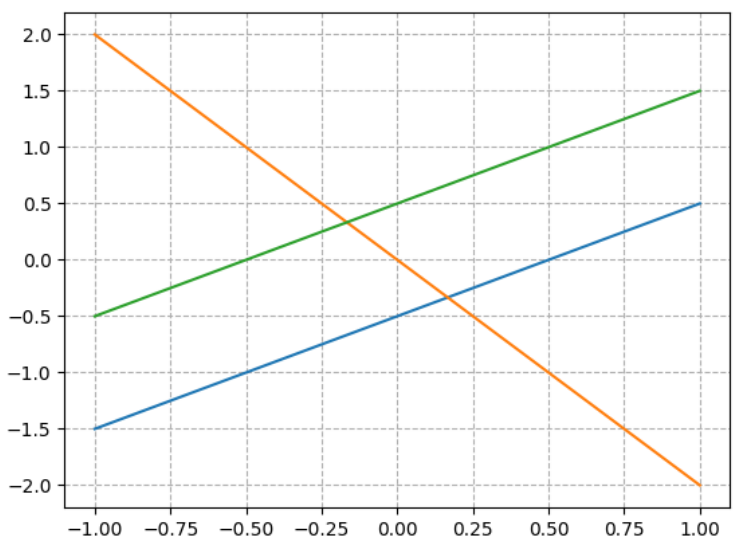
[1]:
```
from LagrangeBasisFuncDerivative import PlotLagrangeBasisDerivatives,␣
↪PlotTwoDBasisFunctionParentDomain
```

Problem 3

Plot the derivative of all one dimensional Lagrange basis functions of degree 2 over the interval $[-1, 1]$ with interpolation points evenly split on this interval

[2]:
```
p = 2            # Degree of polynomial
pts = [-1, 0, 1] # Interpolation points
PlotLagrangeBasisDerivatives(p, pts)
```



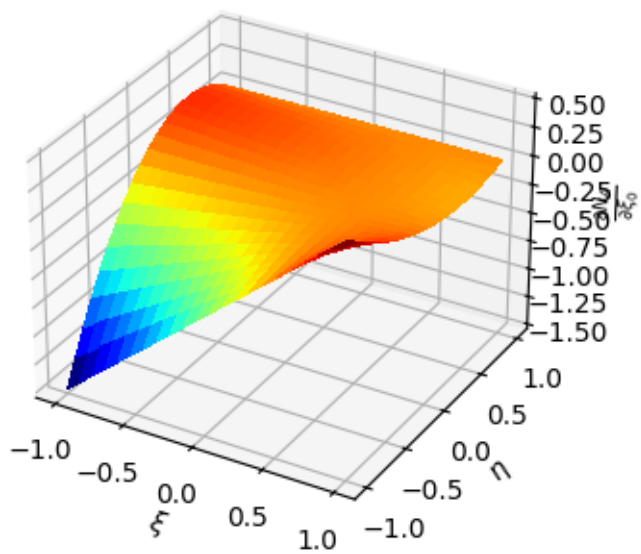
Problem 4

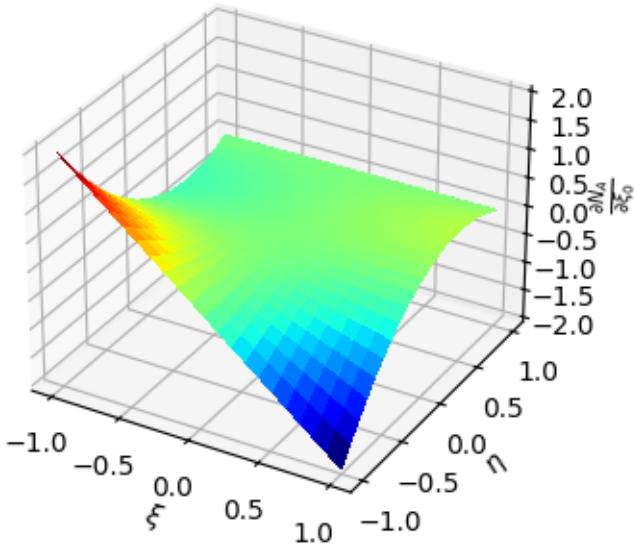
Plot the partial derivative of all two dimensional Lagrange basis functions of degree 2 in both directions, defined over the square $[-1, 1]^2$, with interpolation points evenly distributed in each interval direction. Plot these for derivatives with respect to ξ and derivatives with respect to η.

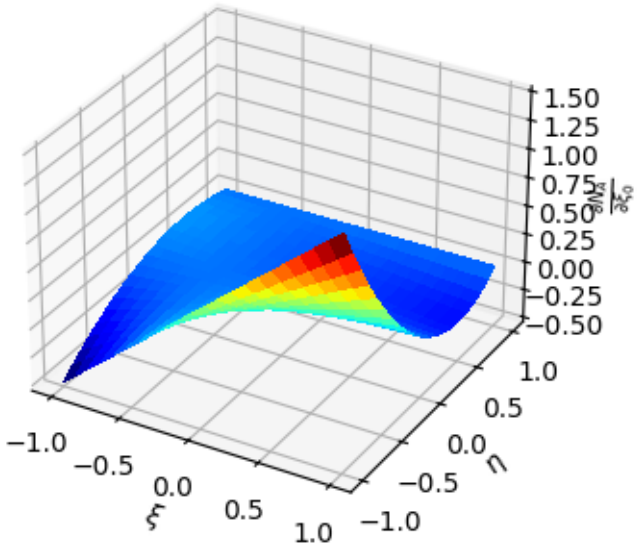
[3]:
```
degs = [2, 2]    # Polynomial degrees in each direction
interp_pts = [
    [-1, 0, 1],
    [-1, 0, 1]
]                # Interpolation points in each direction
```

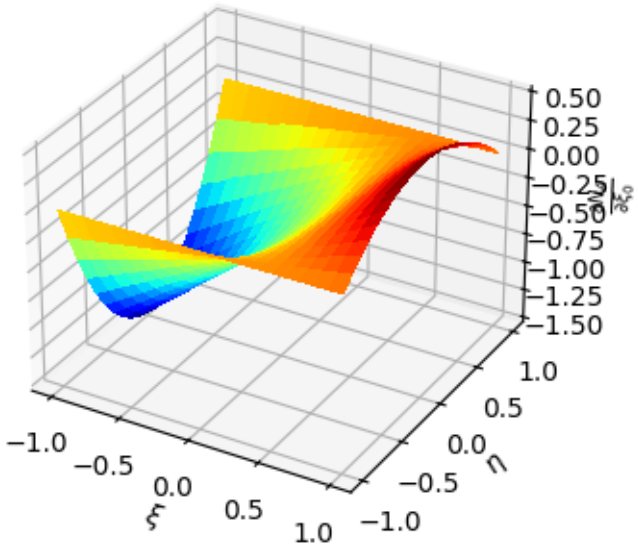
ξ Partial Derivatives

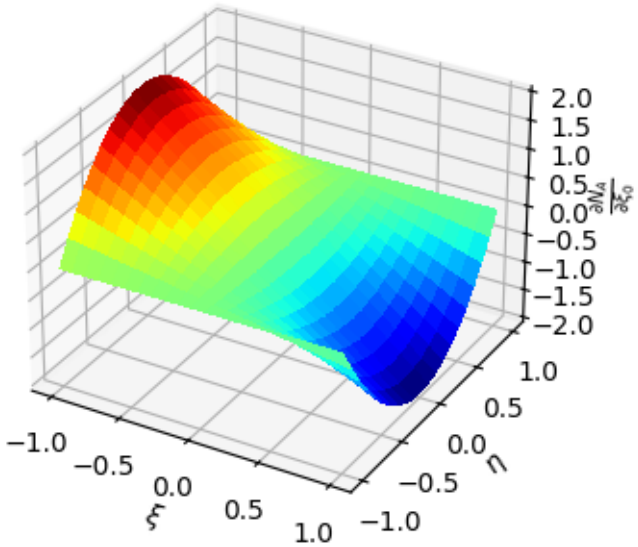
[4]:
```
for A in range(9):
    PlotTwoDBasisFunctionParentDomain(
        A, degs, interp_pts, 0
    )
```

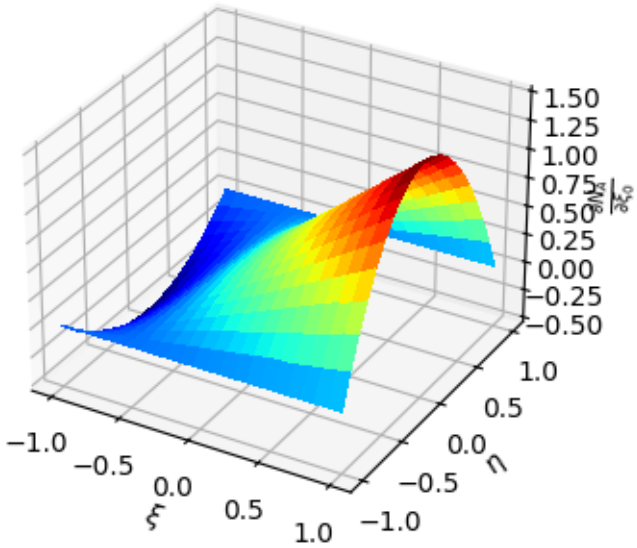


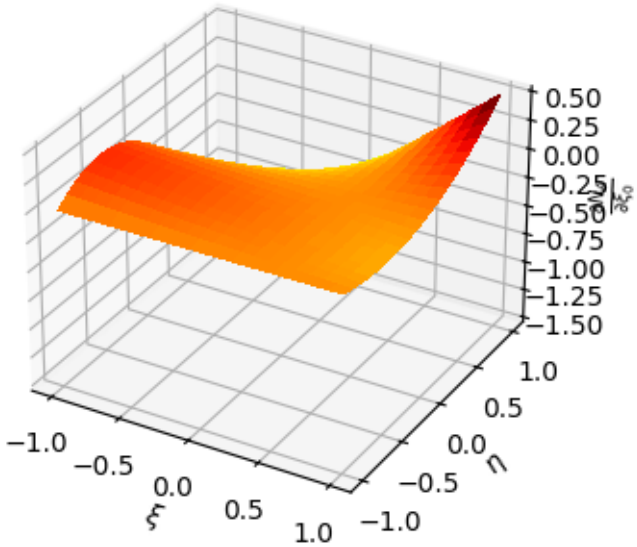

2.0
1.5
1.0
0.5
0.0
−0.5
−1.0
−1.5
−2.0
$\frac{\partial N_A}{\partial \xi_0}$
−1.0
−0.5
0.0
0.5
1.0
ξ
1.0
0.5
0.0
−0.5
−1.0
η

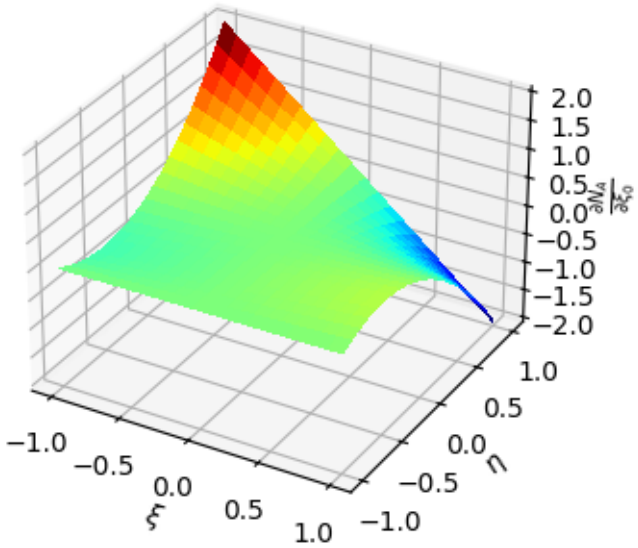

1.50
1.25
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
$\frac{\partial N_B}{\partial \xi_0}$
−1.0
−0.5
0.0
0.5
1.0
ξ
1.0
0.5
0.0
−0.5
−1.0
η

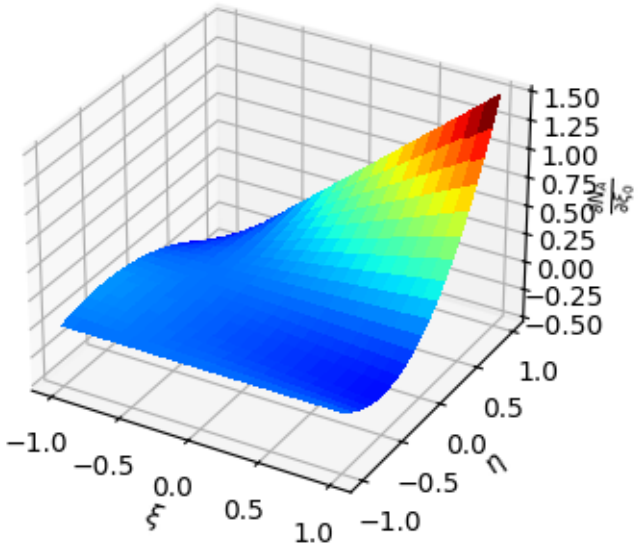

0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00
−1.25
−1.50
$\frac{\partial N_a}{\partial x_0}$
−1.0
−0.5
0.0
0.5
1.0
ξ
1.0
0.5
0.0
−0.5
−1.0
η


2.0
1.5
1.0
0.5
0.0
−0.5
−1.0
−1.5
−2.0
$\frac{\partial N_a}{\partial x_0}$
−1.0
−0.5
0.0
0.5
1.0
ξ
1.0
0.5
0.0
−0.5
−1.0
η


1.50
1.25
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
$\frac{\partial N_\xi}{\partial \xi_0}$
−1.0
−0.5
0.0
0.5
1.0
ξ
1.0
0.5
0.0
−0.5
−1.0
η

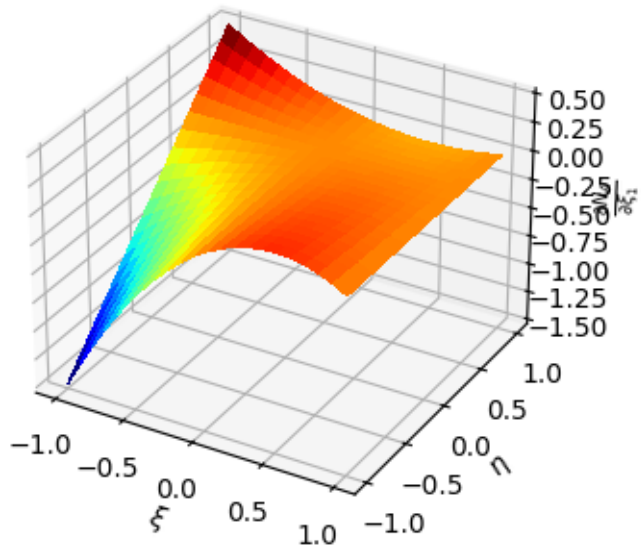

0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00
−1.25
−1.50
$\frac{\partial N_\eta}{\partial \xi_0}$
−1.0
−0.5
0.0
0.5
1.0
ξ
1.0
0.5
0.0
−0.5
−1.0
η

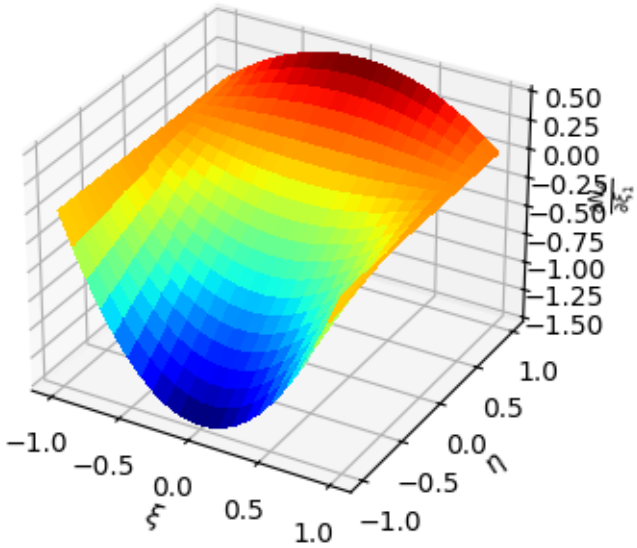

2.0
1.5
1.0
0.5
0.0
−0.5
−1.0
−1.5
−2.0
$\frac{\partial N_A}{\partial \xi_0}$
−1.0
−0.5
0.0
0.5
1.0
ξ
1.0
0.5
0.0
−0.5
−1.0
η

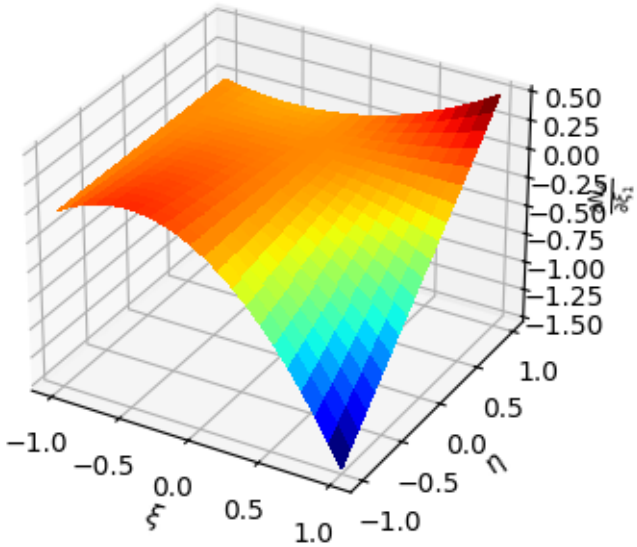

1.50
1.25
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
$\frac{\partial N_B}{\partial \xi_0}$
−1.0
−0.5
0.0
0.5
1.0
ξ
1.0
0.5
0.0
−0.5
−1.0
η

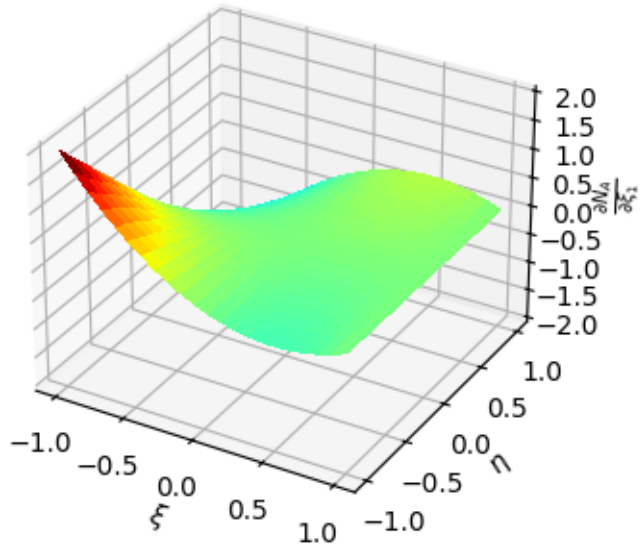
η Partial Derivatives

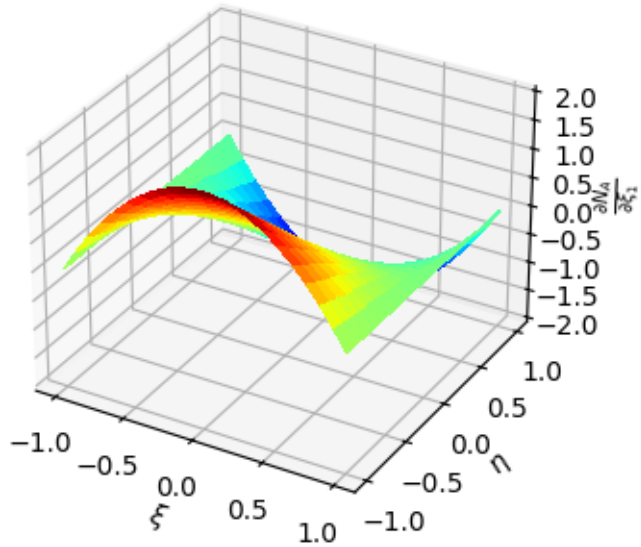
[5]:
```
for A in range(9):
    PlotTwoDBasisFunctionParentDomain(
        A, degs, interp_pts, 1
    )
```

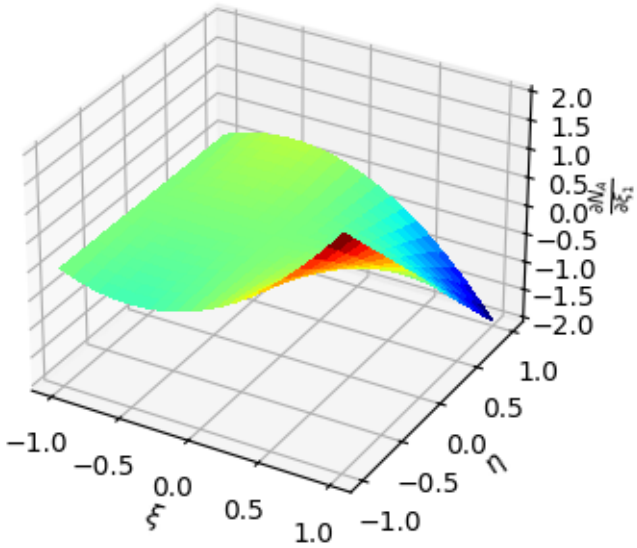


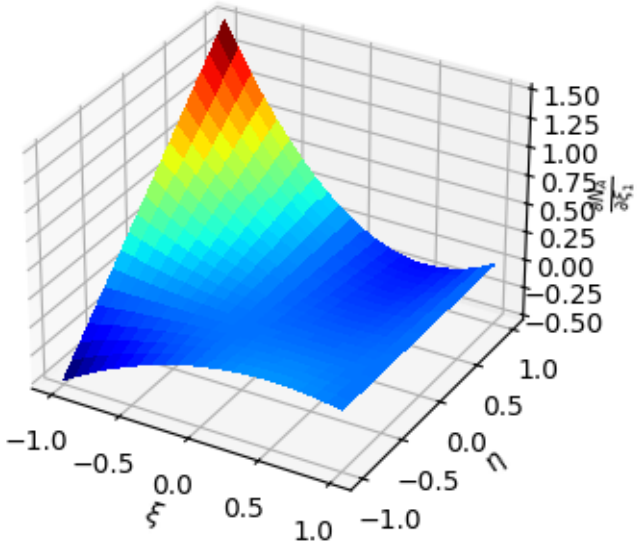

0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00
−1.25
−1.50
$\frac{\partial N_1}{\partial \xi_1}$
−1.0
−0.5
0.0
0.5
1.0
ξ
−1.0
−0.5
0.0
0.5
1.0
η

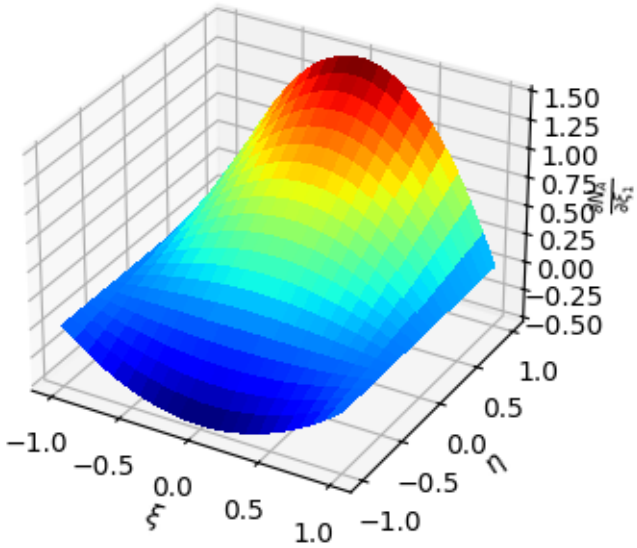

0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00
−1.25
−1.50
$\frac{\partial N_1}{\partial \xi_1}$
−1.0
−0.5
0.0
0.5
1.0
ξ
−1.0
−0.5
0.0
0.5
1.0
η

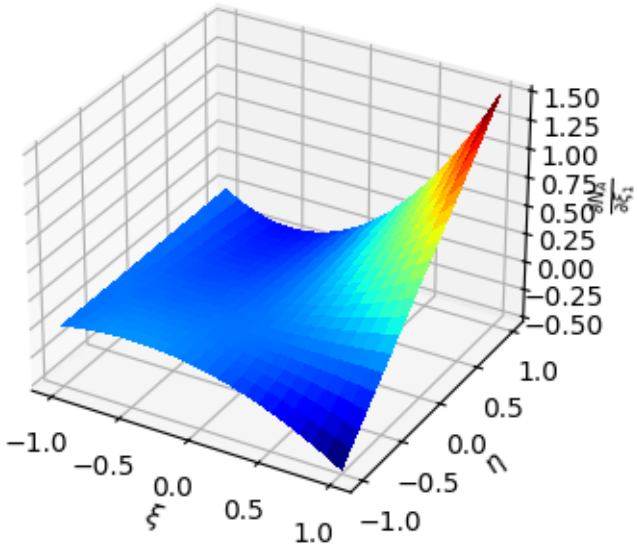

2.0
1.5
1.0
0.5
0.0
−0.5
−1.0
−1.5
−2.0
$\frac{\partial N_A}{\partial \xi_1}$
−1.0
−0.5
0.0
0.5
1.0
ξ
1.0
0.5
0.0
−0.5
−1.0
η


2.0
1.5
1.0
0.5
0.0
−0.5
−1.0
−1.5
−2.0
$\frac{\partial N_A}{\partial \xi_1}$
−1.0
−0.5
0.0
0.5
1.0
ξ
1.0
0.5
0.0
−0.5
−1.0
η


2.0
1.5
1.0
0.5
0.0
−0.5
−1.0
−1.5
−2.0
$\frac{\partial N_A}{\partial \xi_1}$
−1.0
−0.5
0.0
0.5
1.0
ξ
1.0
0.5
0.0
−0.5
−1.0
η


1.50
1.25
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
$\frac{\partial N_B}{\partial \xi_1}$
−1.0
−0.5
0.0
0.5
1.0
ξ
1.0
0.5
0.0
−0.5
−1.0
η


1.50
1.25
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
$\frac{\partial N_8}{\partial \xi_1}$
−1.0
−0.5
0.0
0.5
1.0
ξ
1.0
0.5
0.0
−0.5
−1.0
η


1.50
1.25
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
$\frac{\partial N_8}{\partial \xi_1}$
−1.0
−0.5
0.0
0.5
1.0
ξ
1.0
0.5
0.0
−0.5
−1.0
η